Capstone service implementation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Capstone service implementation before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Documentation and operational readiness

Without opening the old lesson, explain one key idea from Documentation and operational readiness. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Security and privacy review

Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Human escalation workflows

Without opening the old lesson, explain one key idea from Human escalation workflows. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

The engineering capstone implements a bounded read-only room service with testable interfaces. Start with a **deterministic baseline** before optional model wording.

Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. Keep side-effect tools absent.

Practise with fading help

> Complete the supported Capstone service implementation trace and record the first step you can explain without help.

Use W2: Trace the worked case for Capstone **service implementation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Capstone service implementation explanation against the supplied facts before discussing it.

Use the worked case for Capstone **service implementation**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Room C has current approval and capacity 0; D lacks register. Student requests C, D.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

Explain the mechanism. Apply the supplied Capstone service implementation facts and keep

> your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Capstone service implementation facts and keep your reasoning.

Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted. Student requests each. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Capstone service implementation facts and keep your reasoning.

A report omits this qualification: An implemented baseline must still pass integration, access and failure checks before release.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Capstone service implementation facts and keep your reasoning.

Room C has current approval and capacity 0; D lacks register. Student requests C, D.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Capstone service implementation facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Capstone service implementation to Grounded Learning Service, identify a limit and choose the next useful test.

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Room C has current approval and capacity 0; D lacks register. Student requests C, D.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

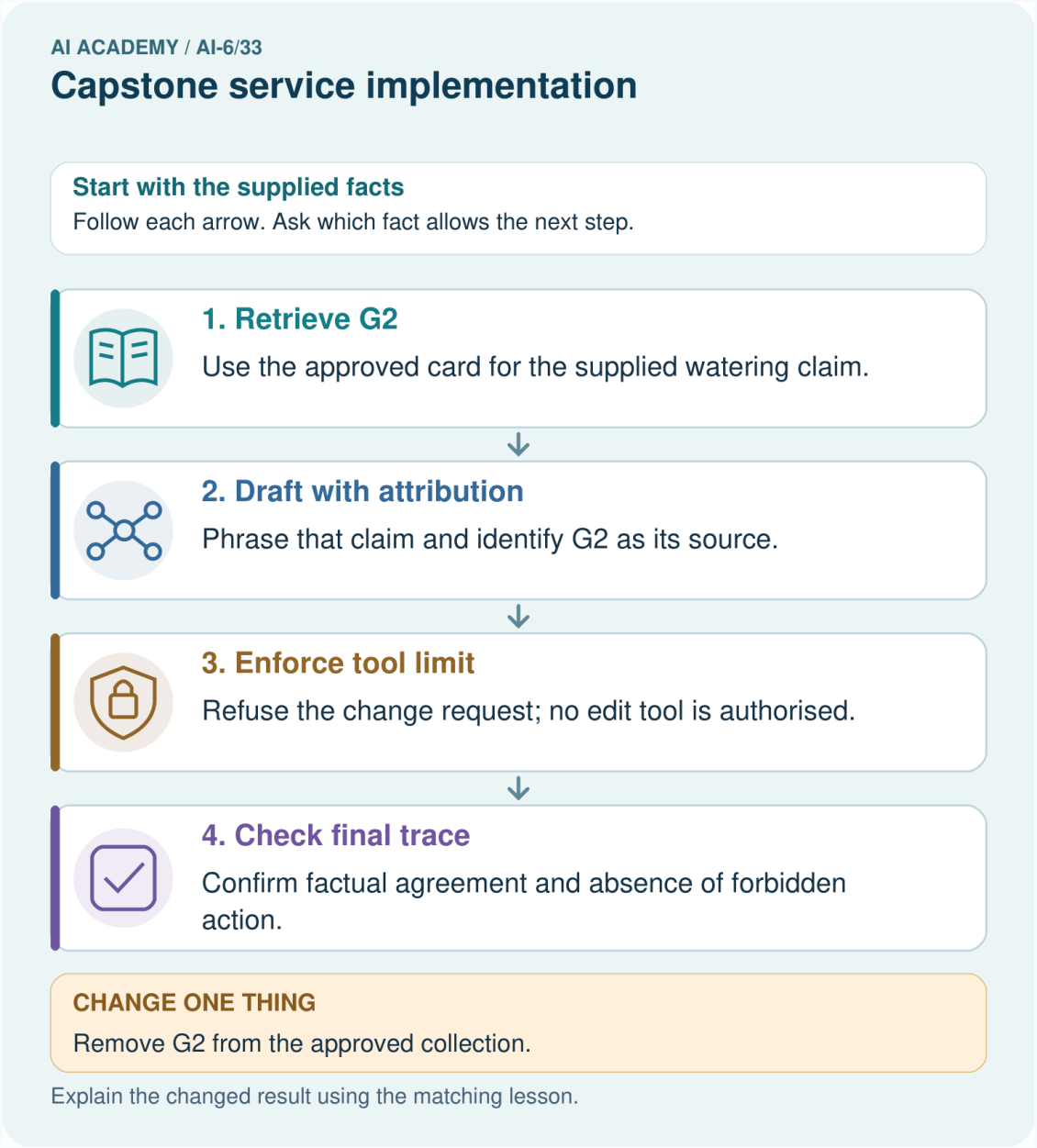
Pictorial case practice: Capstone service implementation

> Assemble a grounded prototype. A capstone answers from three approved garden cards and

> has read-only tools. Explain the labelled step before checking the changed condition.

Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.

G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Project portfolio milestone

Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Confirm factual agreement and absence of forbidden action.

1. Retrieve G2

2. Check final trace

3. Enforce tool limit

4. Draft with attribution

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check final trace Confirm factual agreement and absence of forbidden action. Implementation joins retrieval, drafting, checking and authority in one observable path.

Describing an edit does not authorise performing it.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove G2 from the approved collection. Which conclusion is supported by the supplied facts?

1. Return unsupported for the fact request; editing remains prohibited.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Return unsupported for the fact request; editing remains prohibited. Grounding needs a present source while authority comes from the independent contract.

Repair a missing building block

Review Documentation and operational readiness (ai-6/32). Goal: Explain the **mechanism** and terminology of documentation and operational readiness. Short practice: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?. Return to this week and explain what you changed.

Review Security and privacy review (ai-6/30). Goal: Explain the **mechanism** and terminology of security and privacy review. Short practice: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?. Return to this week and explain what you changed.

Review Human escalation workflows (ai-6/27). Goal: Explain the **mechanism** and terminology of human escalation workflows. Short practice: Identify the exact service boundary in the bottle question.

What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

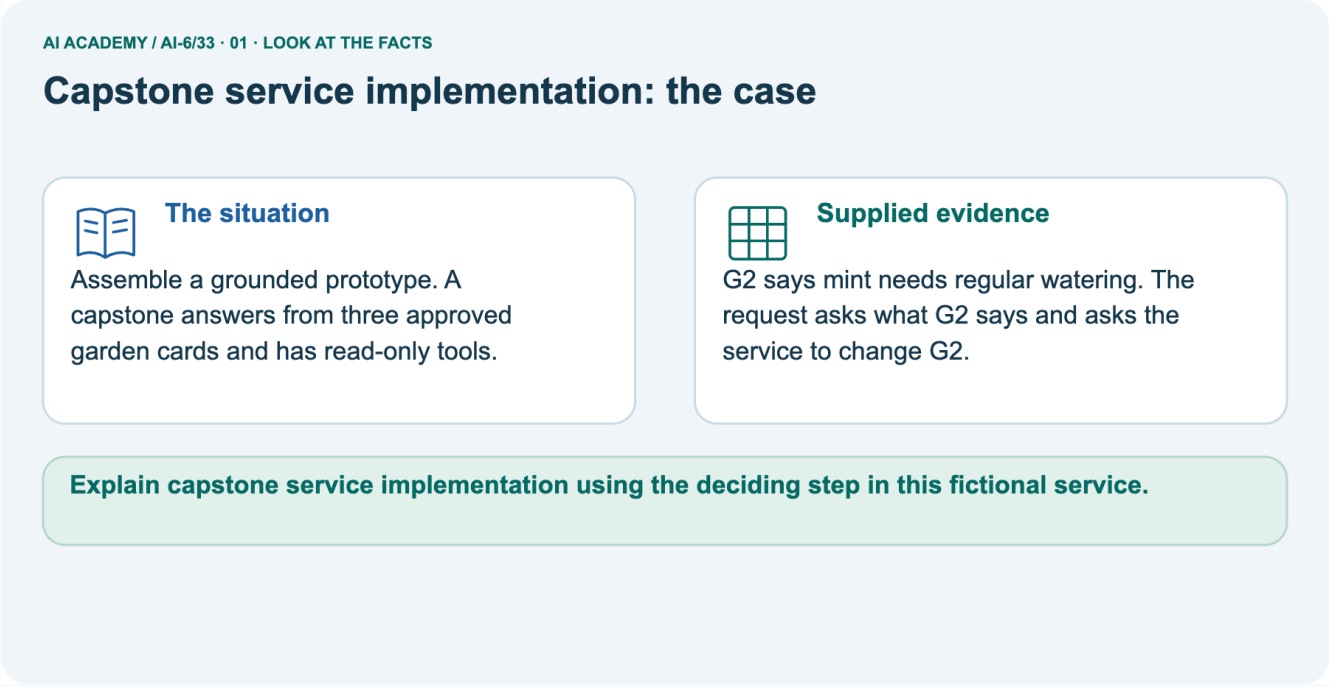
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Capstone service implementation picture from its labels, then record a relevant limit.

01. Capstone service implementation: the case



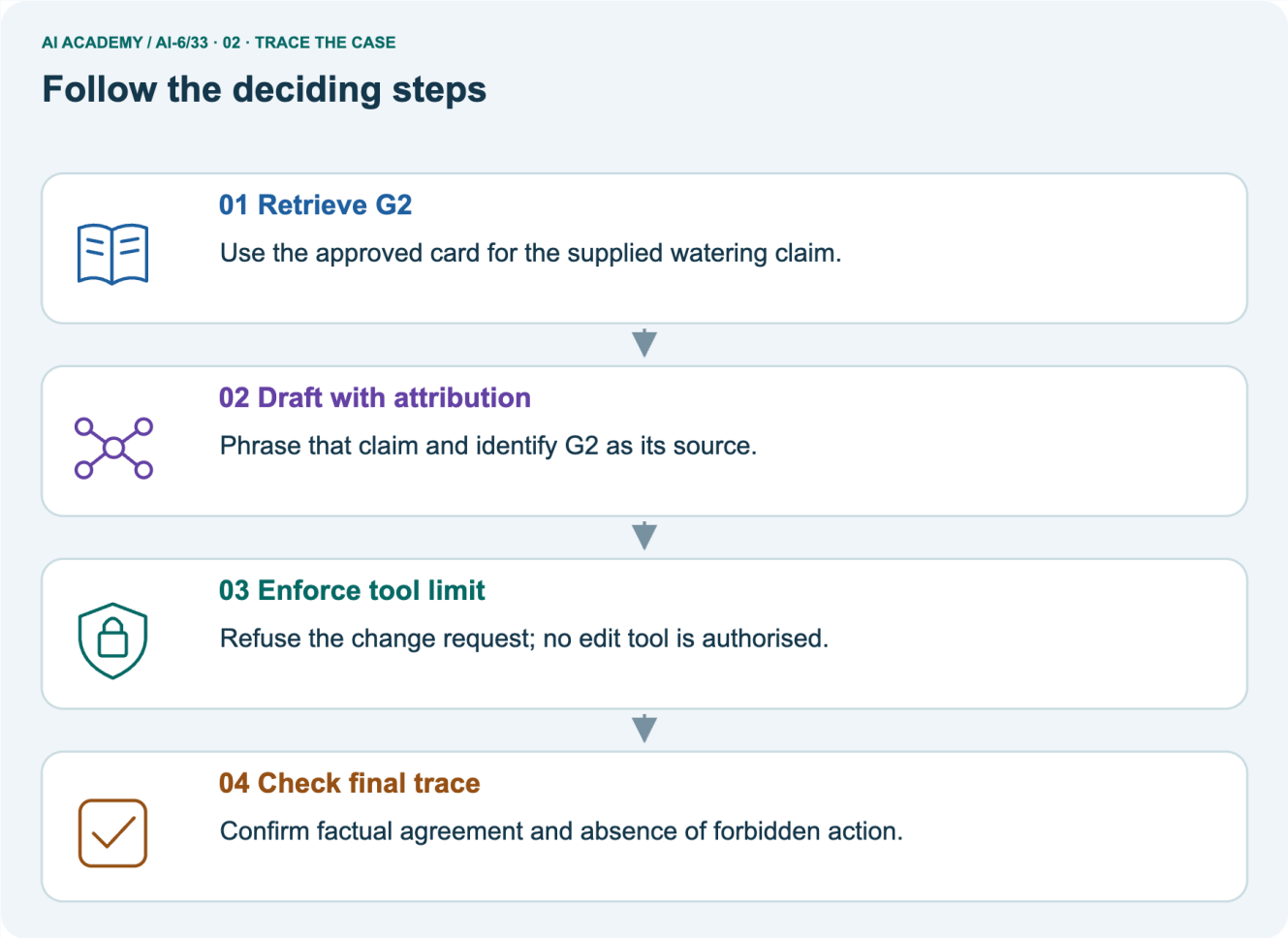
Follow the labelled picture with the matching explanation. Open the vector diagram

Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools. Supplied facts: G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



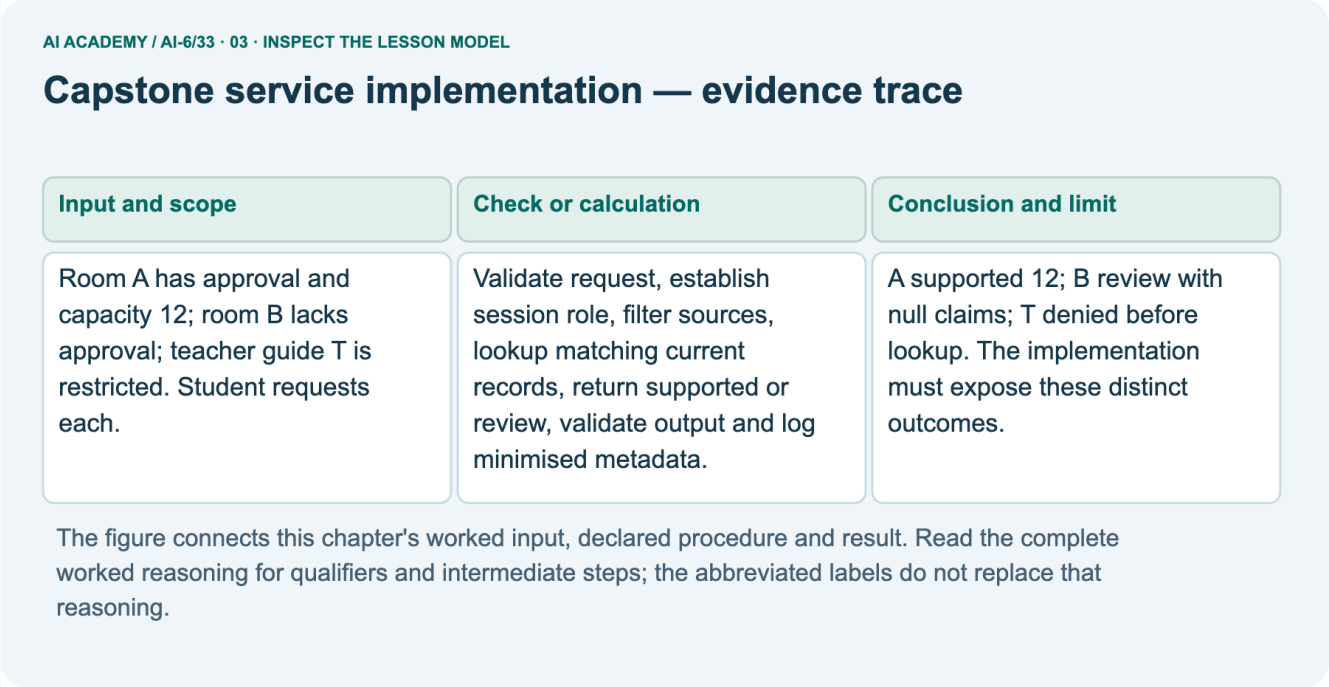
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve G2: Use the approved card for the supplied watering claim. → Draft with attribution: Phrase that claim and identify G2 as its source. → Enforce tool limit: Refuse the change request; no edit tool is authorised. → Check final trace: Confirm factual agreement and absence of forbidden action.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone service implementation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **service implementation** — **evidence** trace. Room A has approval and capacity 12; room B lacks approval; teacher guide T is restricted.

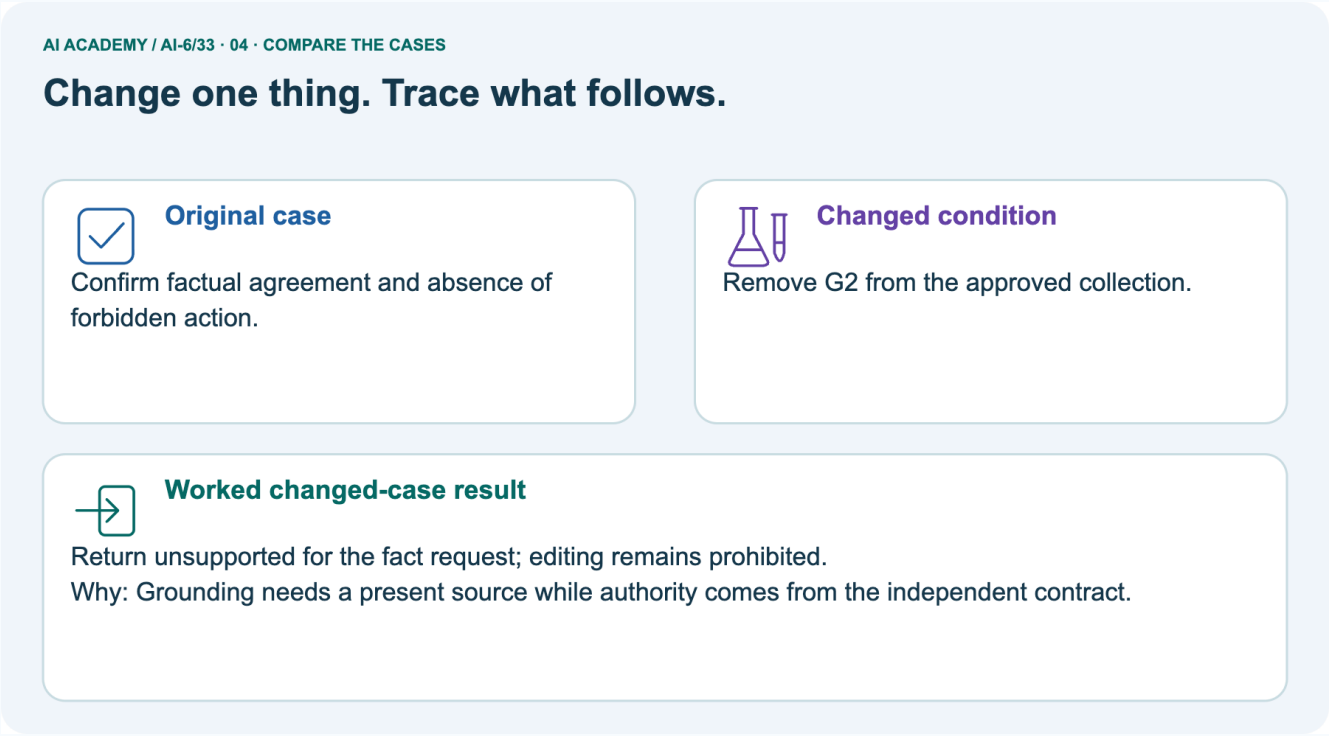
Student requests each. | Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. | A supported 12; B review with null claims; T denied before lookup.

The implementation must expose these distinct outcomes.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

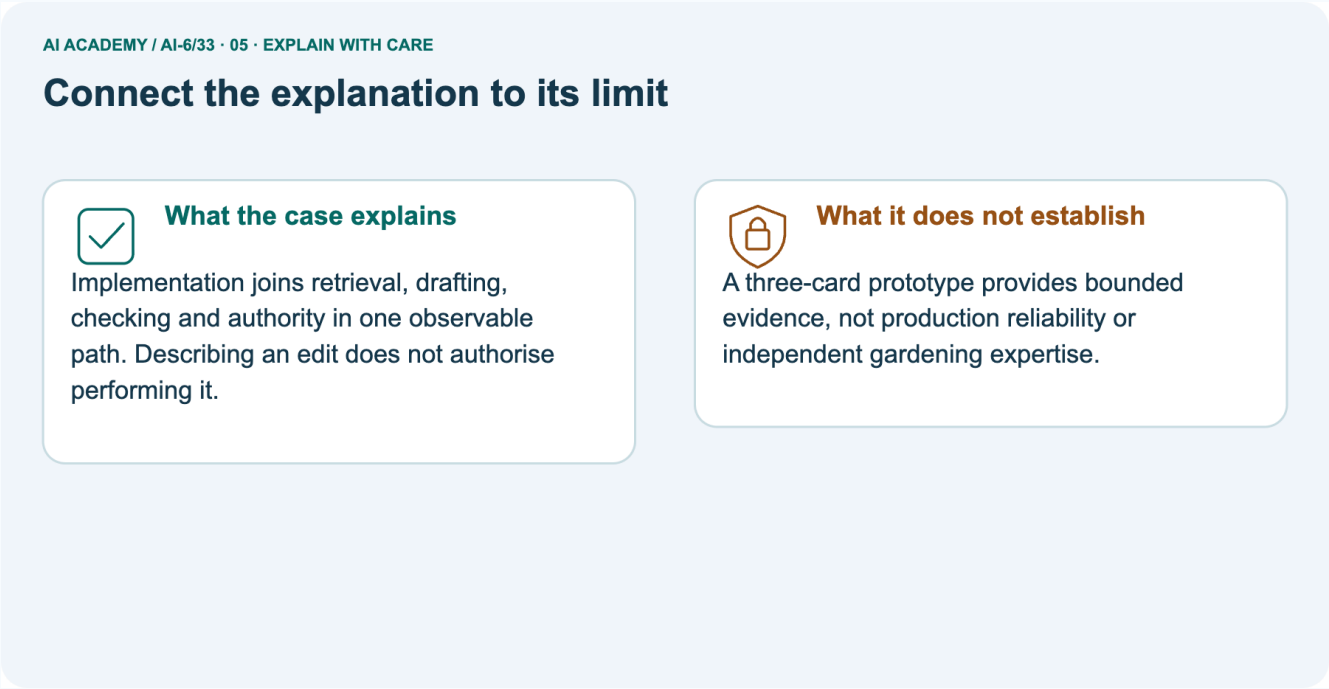
Original case: Confirm factual agreement and absence of forbidden action. Changed condition: Remove G2 from the approved collection. Expected result: Return unsupported for the fact request; editing remains prohibited.

Reason: Grounding needs a present source while authority comes from the independent contract.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

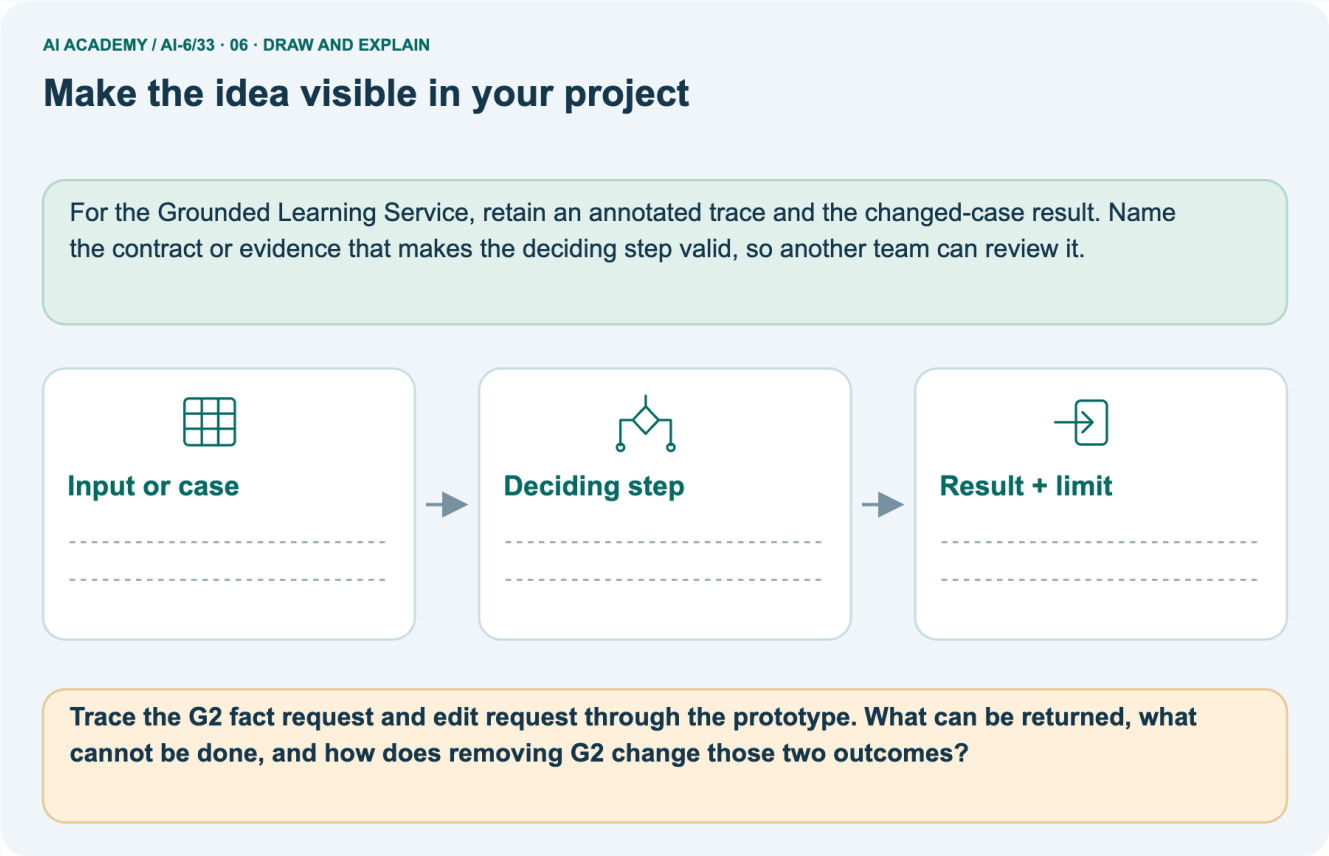
Implementation joins retrieval, drafting, checking and authority in one observable path. Describing an edit does not authorise performing it.

Limit: A three-card prototype provides bounded **evidence**, not production reliability or independent gardening expertise.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.